Replication Documentation

Rural Electrification in the Brazilian Amazon:
No Environment-Development Tradeoff

github.com/lavifi/Rural-Electrification

This PDF combines the repository README, reproduction instructions, codebook, exhibit provenance manifest, and known-issues notes for convenient review. The Markdown files in the repository remain the authoritative editable versions.

Documentation snapshot: October 5, 2026

Repository README

This repository reproduces the quantitative results in Laura Villalobos, Jill Caviglia-Harris, Daniel Harris, and Elaine Frey, “Rural Electrification in the Brazilian Amazon: No Environment-Development Tradeoff,” Land Economics (2026), https://doi.org/10.3368/le.103.1.020926-0023.

Contents

- `data/Villalobos_et_al_2026.dta`: analysis-ready, de-identified property-year panel containing 37 documented variables.
- `code/run.do`: portable entry point for the complete Stata workflow.
- `code/Villalobos et al. 2026.do`: analysis that produces the reported tables and figures.
- `tables/`: machine-generated LaTeX tables plus two documented, externally sourced descriptive tables.
- `figures/main/`: final Figures 1-3, named to match the published paper.
- `figures/appendix/`: final Appendix Figures A1-A5, named to match the published appendix.
- `CODEBOOK.md`: variable definitions, samples, units, and missingness conventions.
- `REPLICATION.md`: software requirements, runtime instructions, and exhibit mapping.
- `MANIFEST.md`: provenance and reproducibility status of each reported exhibit.
- `output/pdf/Replication_Documentation.pdf`: PDF copy of the README, reproduction instructions, codebook, manifest, and known-issues notes.

Quick start

Stata 19 is recommended.

1. Download the repository ZIP from GitHub.
2. Right-click the ZIP and select **Extract All**. Do not open or run `run.do` from inside the compressed ZIP folder, because Windows may use a temporary path that Stata cannot find later.
3. Open Stata.
4. If this is the first run, enter the following in Stata's **Command** window, replacing `FULL-PATH` with the location of the extracted folder:

```
do "FULL-PATH/code/install_stata_packages.do"
```

5. Enter the following as one command in Stata's **Command** window:

```
do "FULL-PATH/code/run.do" "FULL-PATH"
```

For example:

```
do "C:/Users/username/Downloads/Rural-Electrification-main/code/run.do" "C:/Users/username/Downloads/Rural-Electrification-main"
```

The first path identifies the do-file. The second path is not another do-file: it tells the script where the replication-package root is located. That root folder must contain `README.md`, `code`, `data`, `figures`, and `tables`. Opening `run.do` and clicking Stata's Run button without supplying the second path is not sufficient.

The entry point accepts paths containing spaces. It reads the released dataset, overwrites generated tables and figures in place, and writes `logs/analysis.log`. No code edits are required. All analysis is written in Stata; the `.md` files are documentation, not R Markdown programs.

Reproducibility scope

The released Stata file reproduces Tables 1-4, Appendix Tables A3-A5, main Figures 2-3, and Appendix Figures A1, A3, and A4. Main Figure 1, Appendix Figure A2, Appendix Figure A5, and Appendix Tables A1-A2 use GIS or public-source inputs that

are not contained in the analysis dataset; their provenance is documented in `MANIFEST.md`. The package includes their final static files.

Data protection

The released data contain anonymous lot identifiers and no names, addresses, coordinates, or contact information. The identifiers are stable over time so that users can reproduce the panel estimators.

Contact

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Reproduction Instructions

Software

- Stata 19 (tested with Stata/SE 19.5 on Windows).
- User-written Stata commands: `csdid2` (including `csdid2_plot`), `estout`, `distinct`, `unique`, `coefplot`, `pretrends`, `grc1leg`, `reghdfe`, `wildbootstrap`, `boottest`, and `honestdid`.
- A LaTeX distribution is optional and is needed only to typeset the generated `.tex` tables in a separate document.

The analysis stops with a readable error if the dataset or a required command is unavailable. Internet access is not required after the listed Stata packages have been installed.

For a first-time setup, inspect and run `code/install_stata_packages.do` while connected to the internet. The successful verification run used `estout` 3.31, `unique` 1.2.4, `coefplot` 1.8.8, `pretrends` 0.5.1, `grc1leg` 1.0.5, `reghdfe` 6.12.5, and `honestdid` 1.3.0.

Run

If the repository was downloaded as a ZIP file, first right-click the ZIP and select **Extract All**. Do not run the code from inside the compressed ZIP folder: Windows may open it from a temporary location that Stata cannot find later.

Open Stata and enter the following as one command in the **Command** window:

```
do "FULL-PATH/code/run.do" "FULL-PATH"
```

Replace both instances of `FULL-PATH` with the absolute path to the extracted replication folder. For example:

```
do "C:/Users/username/Downloads/Rural-Electrification-main/code/run.do" "C:/Users/username/Downloads/Rural-Electrification-main"
```

The first quoted path identifies `run.do`. The second quoted path is the script's root-directory argument, not another do-file. It must identify the folder containing `README.md`, `code`, `data`, `figures`, and `tables`. Opening `run.do` in the Do-file Editor and clicking Run without passing that argument will produce an error.

The master script uses a fixed bootstrap seed (`123456789`) and overwrites generated outputs. A text log is written to `logs/analysis.log`.

Expected runtime depends on hardware. The land-use panel contains 343,332 property-year observations, and repeated bootstrap inference is the slowest stage.

Expected samples

- Full released panel: 343,332 observations, 9,537 properties, 1985-2020.
- Main land-use sample: 304,920 observations, 8,470 properties, 36 years.
- Socioeconomic panel sample: 468 observations from 156 properties across the 1996, 2000, and 2005 survey waves; total income is nonmissing for the 458 rows used in Table 2.
- Socioeconomic estimation sample: 330 household-year observations from 110 households not treated in 1996 and observed at least twice.

Output mapping

Final figure filenames follow the numbering in the published paper:

Article exhibit	Generated or supplied file
Table 1	`tables/table1.tex`
Table 2	`tables/table2.tex`

Article exhibit	Generated or supplied file
Table 3	`tables/table3.tex`
Table 4	`tables/table4.tex`
Figure 1	`figures/main/figure1_study_area.jpg` (supplied GIS output)
Figure 2	`figures/main/figure2_deforestation_event_study.pdf`
Figure 3	`figures/main/figure3_socioeconomic_event_studies.png`
Appendix Table A1	`tables/t_des_census.tex` (supplied public-source table)
Appendix Table A2	`tables/t_interpolation.tex` (supplied GIS-method table)
Appendix Table A3	`tables/tableS3.tex`
Appendix Table A4	`tables/tableS4.tex`
Appendix Table A5	`tables/tableS5.tex`
Appendix Figure A1	`figures/appendix/figureA1_summary_ATT.png`
Appendix Figure A2	`figures/appendix/figureA2_grid_connection_map.jpg` (supplied GIS output)
Appendix Figure A3	`figures/appendix/figureA3_deforestation_sensitivity.png`
Appendix Figure A4	`figures/appendix/figureA4_socioeconomic_sensitivity.png`
Appendix Figure A5	`figures/appendix/figureA5_GDP_comparison.png` (supplied World Bank/OECD chart)

Codebook

Dataset overview

`data/Villalobos_et_al_2026.dta` is an unbalanced property-year panel for rural properties in Greater Ouro Preto do Oeste, Rondônia, Brazil (`studycode == 2` in the source survey panel). It contains 343,332 observations for 9,537 anonymous lots over 1985-2020. The unit of observation is a lot-year.

The source construction retains years 1985-2020 and excludes lots occupied by renters (`type == 3` in any observed wave). The released data contain no names, addresses, coordinates, or contact details.

Coding and missing values

- A dot (`.`) is Stata numeric missing.
- Electricity indicators use `0 = not connected` and `1 = connected`.
- Sample indicators use `1 = included`; observations outside the indicated sample are missing, not zero.
- Survey variables are structurally missing outside survey observations.
- For interviewed households in 1996, 2000, 2005, 2009, or 2019, missing activity-income values were coded as zero because they indicate no income from that activity.
- Appliances not included in early questionnaires were coded as zero for interviewed households. These are assumed zeros because survey design and piloting indicated that the appliances were not present or materially relevant, rather than directly observed zeros.
- The released `.dta` contains no Stata value-label definitions.

Monetary variables

All released income variables are annual gross revenue in thousands of 2019 Brazilian reais (R$); production costs are not deducted. Milk and beef revenue and crop revenue are derived from on-property activities. Off-property revenue is labor income from activities such as hourly or part-time work on another property, retail work, commercial sales, or other professions.

The historical income variables used in the paper's 1996, 2000, and 2005 socioeconomic analysis were reported in real values and treated by the historical construction workflow as constant year-2000 reais. The construction code converts them to constant year-2019 reais using the Brazilian consumer price index (`FP.CPI.TOTL`, World Development Indicators), multiplying by `CPI_2019 / CPI_2000`, and then divides by 1,000. The 2019 survey wave is not part of the paper's socioeconomic analysis.

Government payments combine pensions/social-security income, Bolsa Escola, and Bolsa Família. The paper describes the latter programs collectively as Brazilian cash transfers.

Variable definitions

Identifiers and time

Variable	Definition and coding
`id`	Anonymous lot identifier, stable across years. Household-lot matches across survey waves were based on the same identifier and manually checked.
`year`	Calendar year, 1985-2020.
`sett_year`	Year in which the settlement was established.

Variable	Definition and coding
year_se	Equally spaced socioeconomic-analysis time index used with csdid2: survey years 1996, 2000, 2005, and 2009 are coded 1996, 2000, 2004, and 2008, respectively. The original survey years have irregular four- and five-year gaps, which prevented the command from identifying the appropriate pre-treatment comparison periods for the dynamic-effects analysis. Following guidance from the command's author, the recoding makes every transition between successive survey waves one regular four-unit interval and aligns the time index with the cohort codes in gvar_se.

Income outcomes

Variable	Definition and construction
incmilk	Annual income from on-property milk production, thousands of 2019 R$.
incbeef	Annual income from on-property beef production, thousands of 2019 R$. Beef income was not collected in the earliest survey and is coded zero for interviewed households when missing.
inccrop	Annual crop income, thousands of 2019 R$: annual-crop income plus perennial-crop income.
incoff	Annual off-property labor income, thousands of 2019 R$.
incgov	Annual government payments, thousands of 2019 R$: pensions/social security plus Bolsa Escola and Bolsa Família.
incometot	Total annual income, thousands of 2019 R$: incmilk + incbeef + inccrop + incoff + incgov.

The component-sum identity for incometot holds up to floating-point rounding in the released data.

Appliances

Variable	Definition and coding
fridge	Number of refrigerators and freezers owned by the household.
fans	Number of fans owned by the household.
washers	Number of washing machines owned by the household. Not included in some early questionnaires; assumed zero for interviewed households when unrecorded.
microwaves	Number of microwaves owned by the household. Not included in some early questionnaires; assumed zero for interviewed households when unrecorded.
milktanks	Number of milk tanks owned by the household. Not included in some early questionnaires; assumed zero for interviewed households when unrecorded.
tvs	Number of televisions owned by the household.
phones	Number of landline telephones owned by the household.
cell	Number of cell phones owned by the household. Not included in early questionnaires; assumed zero for interviewed households when unrecorded.
durelecwork	Work-appliance count: fridge + fans + washers + microwaves + milktanks.
durelecplay	Leisure-appliance count: tvs + phones + cell.

The two appliance-index identities hold exactly in the released data.

Livestock and milk production

Variable	Definition and construction
cattlelot	Total number of cattle owned on the lot.
cattlemilk	Number of milk cows.

Variable	Definition and construction
tmilkharv	Sum of dry-season and wet-season daily milk harvest, liters/day. An observed seasonal value is retained when the other seasonal value is missing; the result is missing only when both seasonal values are missing. In s_socio, both seasonal values are missing for 10 observations (5 in 1996 and 5 in 2005).
milkyield	Total daily milk harvest divided by the number of milk cows, liters/day/cow. A zero denominator produces missing.
price_milk	Simple, unweighted mean of the dry- and wet-season milk prices, expressed in 2019 R$/liter. Both seasonal prices were asked in the survey. Following the paper's analysis code, each seasonal input price is multiplied by the common CPI_2019 / CPI_2000 factor before the mean is calculated.
cattle_ha	Total cattle (cattlelot) divided by household-reported pasture area in hectares. A zero pasture denominator produces missing.

Land use

The land-cover source is **MapBiomas Version 6: SDSU Filtered Product - OPO lots**. It provides annual 30-meter land-cover measurements for the study properties over 1985-2020.

Variable	Definition and construction
dMBdefor	Annual change in MapBiomas forest area, hectares: forest area in year t-1 minus forest area in year t. Positive values represent measured forest loss; negative values represent measured forest gain or year-to-year classification variability. The value is missing for the first observed year of each lot or when years are not consecutive.

The MapBiomas lot-area check sums forest, savanna, wetland, grassland, pasture, urban, water, soybean, other, and cloud classes. Lots whose calculated area exceeds 600 hectares are excluded from the land-use analysis.

Electricity treatment

Variable	Definition and coding
elecyear	Raw self-reported calendar year in which the household was connected to the electricity grid. It is recorded only on the relevant survey observation and is otherwise missing; it is not repeated across all annual records for a lot.
electricity_year_round	Rounded interpolated grid-connection year, repeated across the annual records for a lot when available. Interpolated values are rounded to the nearest integer using floor(value + 0.5). This released field is identical to gvar_lu_imp wherever observed.
elecyear2_dum	Combined grid-treatment indicator. Equals one beginning in the calendar year after the combined connection year.
electricity_year_round_dum	Interpolation-only grid-treatment indicator. Equals one beginning in the calendar year after the rounded interpolated connection year.
gvar_lu	Combined grid-connection year for the preferred land-use specification. The self-reported connection year is used when available; otherwise the rounded interpolated year is used.
gvar_lu_imp	Rounded interpolated grid-connection year used for the interpolation-only robustness specification. Interpolated years are rounded to the nearest integer using floor(value + 0.5).
gvar_se	Socioeconomic treatment cohort on the equally spaced survey index: connection before 1996 → 1996; 1996-1999 → 2000; 2000-2004 → 2004; 2005-2008 → 2008; 2009 or later → 2012.

Because outcomes are annual and connection may occur at any point within a year, treatment begins in the year after the recorded connection year.

Analysis samples

Variable	Definition and coding
`s_socio`	Equals one for observations in 1996, 2000, or 2005 belonging to households interviewed in at least two of those waves. The panel matches use the same anonymous lot identifier and were manually checked. The 2009 wave is omitted because about 96% of properties had electricity by then, leaving too few untreated observations for a credible control group.
`sluilap`	Equals one for the preferred land-use sample: private properties with MapBiomas-calculated lot area no greater than 600 hectares in settlements established before 2000.
`sluilsr`	Equals one for the self-reported-treatment-year robustness sample: properties satisfying `sluilap` that also have a self-reported connection year.

Exhibit Provenance Manifest

Fully reproduced from the released dataset

The master Stata script regenerates article Tables 1-4; Appendix Tables A3-A5; article Figures 2-3; and Appendix Figures A1, A3, and A4. Stata also creates intermediate `.gph` files during execution; these are excluded from the public release because the final numbered figures are supplied in standard image or PDF formats.

Supplied static outputs

- Article Figure 1 (`figures/main/figure1_study_area.jpg`) is a GIS study-area map assembled from settlement, road, municipal, and urban-hierarchy layers.
- Appendix Figure A2 (`figures/appendix/figureA2_grid_connection_map.jpg`) is the GIS interpolation map of connection timing.
- Appendix Figure A5 (`figures/appendix/figureA5_GDP_comparison.png`) is based on World Bank national accounts and OECD National Accounts data.
- Appendix Table A1 (`tables/t_des_census.tex`) uses IBGE SIDRA tables 200, 5938, 3939, 94, and 74; the manuscript notes substitutions of 1997 population and 1999 GDP where 1996 values were unavailable.
- Appendix Table A2 (`tables/t_interpolation.tex`) records GIS interpolation parameters assembled from INCRA records, program timing, and survey reports.

These exhibits are included for completeness but cannot be independently regenerated from the released `.dta` file alone. A future expanded archive should add the source GIS project/layers and the public-data download/processing scripts if redistribution terms allow.

Known Issues and Interpretation Notes

Table 2 property count

The released panel has 156 distinct properties in the socioeconomic sample (`s_socio == 1`) and 468 property-year rows. Total income is observed in 458 rows: 151 in 1996, 156 in 2000, and 151 in 2005.

The published Table 2 reports 151 under “Number of properties” in its “All years” column. This is reproduced by the historical analysis code, which tags the first observation for each property and then requires nonmissing total income on that tagged row. Since the first row is the 1996 observation, the statistic equals the number with nonmissing 1996 income, not the number of distinct properties represented over all survey years. The correct distinct-property count across the socioeconomic sample is 156. This discrepancy affects only the descriptive count; it does not alter any estimate or analysis sample.

Static exhibits

The original packet does not include the raw GIS layers/project or public-data acquisition scripts needed to regenerate article Figure 1, Appendix Figure A2, Appendix Figure A5, or Appendix Tables A1-A2. The final static exhibits and their source notes are supplied. See `MANIFEST.md`.